

# Weather

Current conditions, 48-hour hourly, 10-day, and animated radar. No ads, no trackers, no analytics, no cookie banner, no account, no API keys.

Two builds, same data and same look:

|            | Web PWA                                         | Native Android                    |
|------------|-------------------------------------------------|-----------------------------------|
| Where      | repo root, served by GitHub Pages               | android/                          |
| Built with | one HTML + one CSS + one JS file, no build step | Kotlin, Jetpack Compose, osmdroid |
| Install    | Chrome → Add to Home screen                     | sideload <code>weather.apk</code> |
| Offline    | service worker + localStorage                   | SharedPreferences cache           |

Accuracy is ballpark by design — this is a "do I need a jacket / is that cell going to hit Veenker" app, not a life-safety product. Watches and warnings come straight from the National Weather Service and are the one thing here worth trusting.

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## What's on the screen

| Section | What it shows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alerts  | Active NWS watches/warnings for the exact point. Tap one to read the full text. Hidden when there are none.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Current | Temp, condition, feels-like, today's high/low, wind + direction, gust, humidity, UV, sunrise/sunset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Hourly  | Next 48 hours. Horizontal scroll. Blue bar under each hour = chance of precip.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 10 Day  | Day and date, icon, max chance of precip, and a low→high bar scaled across the days shown. <b>Tap a day</b> to swap the hourly strip to that day; tap again for the rolling 48 h. The <code>5 day / 10 day</code> button in the card header switches the length, and the choice is remembered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Radar   | 10 observed NEXRAD frames (last 90 min, 10-minute steps) plus up to 12 HRRR model frames (~3 hr ahead, 15-minute steps), all on the NWS color scale. Opens <b>parked on the newest observed frame</b> — press play to run the loop. Every frame's tiles are fetched as soon as the radar loads, so the first loop draws. Drag the slider to scrub, <code>Expand</code> for fullscreen, pinch to zoom. A <code>+</code> on the timestamp means it's a forecast frame, not an observation. <b>Refresh</b> : frames and storm cells are re-fetched every 5 minutes while the app is on screen, at once on coming back to it if the last load is over 5 minutes old, and with the refresh button; nothing runs in the background. An unchanged frame list leaves the loop alone. A changed one is rebuilt: parked on the newest observed frame stays on the new newest one, otherwise it keeps the frame nearest in time, and a running loop keeps running. A failed refresh leaves the last frames and tracks on screen. <b>Storm tracks</b> : moving cells ( $\geq 5$ kt, $\geq 40$ dBZ) within 300 km get a dot, a line to where they'll be in 60 minutes, ticks at 15 / 30 / 45 min and an arrowhead — red when the radar flags a tornado vortex signature or a $\geq 50\%$ probability of severe hail. If a track passes within 8 km of the place, a line under the timeline says which cell and roughly when ( <code>Cell 25 mi W, 46 mph, reaches here ≈ 3:02pm</code> ). US only — elsewhere the timestamp reads <code>US only</code> . |

Location: tap the place name to search any city, or the crosshair/pin to use GPS. The last six places you looked at stay as chips in the search screen. Defaults to Ames, IA on a fresh install.

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## Install the Android app

**On your phone**, in Chrome:

1. Open <https://postmaster87.github.io/matts-weather-app/weather.apk>
2. Chrome will warn about the file type — accept the download.
3. Tap the downloaded file. Android will ask to allow installs from Chrome; turn it on, then tap Install.
4. The launcher icon is the sun-behind-cloud, labelled **Weather**.

It is signed with a personal **release** key (4096-bit RSA, CN=Matt), not the Android debug key. Android still calls it an app "from an unknown source" — that is what sideloading looks like — but every later build signed with the same key installs straight over this one and keeps its data. No uninstall step, ever again.

The published `weather.apk` is byte-for-byte the build running on the phone:

```
version 1.3 (versionCode 4)
sha256 50b1111fd42f082c4f4aafafbf737a4335978f8475bd7846a143a4da4846393
```

**The key is two files, neither of them in this repo, and neither recoverable:**

```
%USERPROFILE%\androidkeys\matts-weather-release.p12
android/keystore.properties          (holds the keystore password)
```

Back both up somewhere off the build machine. Lose either one and no future build can update an installed copy — the only path left is uninstall and reinstall, which loses the saved places and the cached forecast.

## Install the web app instead

**On your phone**, in Chrome:

1. Open <https://postmaster87.github.io/matts-weather-app/>
2. Menu (:) → **Add to Home screen**

Both keep the last forecast on the device and open offline with an orange **Offline — cached 2:14pm** line at the bottom instead of pretending the numbers are current.

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## Building the Android app

Everything needed is already on the Windows box: JDK 17, Android SDK 35, build-tools, and Gradle 8.9.

```
cd android && ./gradlew assembleRelease
```

Output lands at `android/app/build/outputs/apk/release/app-release.apk` (~6.8 MB). Copy it to the repo root as `weather.apk` to publish it through Pages.

Two files are needed and neither is committed:

| File                                     | Holds                                                                  |
|------------------------------------------|------------------------------------------------------------------------|
| <code>android/local.properties</code>    | <code>sdk.dir=&lt;path to the Android SDK&gt;</code>                   |
| <code>android/keystore.properties</code> | <code>storeFile, storePassword, keyAlias (weather), keyPassword</code> |

Without `keystore.properties` the release build comes out **unsigned** rather than quietly falling back to the

debug key — Gradle prints a warning saying so, and an unsigned APK refuses to install. That is a much louder failure than one signed with the wrong key, which would only surface later as an update that cannot be applied.

To run it on the emulator:

```
adb install -r android/app/build/outputs/apk/release/app-release.apk
```

## Layout

|                                             |                                                                   |
|---------------------------------------------|-------------------------------------------------------------------|
| android/app/src/main/java/com/matt/weather/ |                                                                   |
| MainActivity.kt                             | activity, permission request, theme                               |
| WeatherVm.kt                                | state, refresh + retry, GPS, search                               |
| data/                                       |                                                                   |
| Models.kt                                   | Place, Current, Hour, Day, Forecast, Alert, RadarFrame, StormCell |
| WeatherApi.kt                               | Open-Meteo, NWS, IEM radar + storm cells, geocoding               |
| StormTracks.kt                              | storm-track math: filter, de-dupe, extrapolate, arrival           |
| Net.kt                                      | URLConnection with a real User-Agent                              |
| Store.kt                                    | SharedPreferences: place, recents, cached payload                 |
| Loc.kt                                      | LocationManager fix + platform reverse geocode                    |
| Fmt.kt                                      | wall-clock string formatting                                      |
| Wmo.kt                                      | WMO code -> icon + label                                          |
| ui/                                         |                                                                   |
| WeatherScreen.kt                            | screen, top bar, fullscreen radar                                 |
| Cards.kt                                    | current / hourly / daily / alert cards                            |
| RadarCard.kt                                | osmdroid map, frame overlays, storm tracks, timeline              |
| WeatherIcon.kt                              | the 12 condition icons, drawn on Canvas                           |
| Theme.kt                                    | palette                                                           |

Icons are drawn on a Compose `Canvas` with the same 24-unit geometry as the web build's SVGs, so both versions look identical rather than merely similar.

## Data sources

| Source                                                   | Used for                                                                                                                          | Key needed |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------|
| <a href="#">Open-Meteo</a>                               | Forecast (hourly + daily)                                                                                                         | No         |
| <a href="#">Open-Meteo Geocoding</a>                     | Place search                                                                                                                      | No         |
| <a href="#">api.weather.gov</a>                          | NWS alerts (US only)                                                                                                              | No         |
| <a href="#">Iowa Environmental Mesonet (IEM)</a>         | Radar tiles: NEXRAD US composite (past) and HRRR simulated reflectivity (future); NEXRAD storm attributes (storm tracks). US only | No         |
| <a href="#">Esri dark canvas</a>                         | Radar basemap                                                                                                                     | No         |
| <a href="#">Leaflet 1.9.4</a> / <a href="#">osmdroid</a> | Map (web / Android)                                                                                                               | No         |

Nothing leaves the phone except the coordinates of the place being looked up. No server, no database, no analytics.

The Android build sends `MattsWeather/1.0 (Android; +<repo URL>)` as its User-Agent because `api.weather.gov` asks callers to identify themselves and throttles generic agents. It deliberately carries **no contact email** — add one there if you ever want NWS to be able to reach you about traffic.

The web build reverse-geocodes the GPS button's label through [BigDataCloud](#); the Android build uses the platform `Geocoder` instead, so it makes no third-party call for that.

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## Known limits

- **Radar is US-only.** IEM's composite and HRRR cover the lower 48 and a margin around it (lat 21-53, lon -130 to -60). Outside that box the radar timestamp reads `US only` and no radar or storm data is fetched. Map zoom is capped at 10.
  - **Forecast radar frames are HRRR model output, not observation.** They come in 15-minute steps (the observed frames are 10 minutes apart), and the model run behind them is usually 1-3 hours old.
  - **Storm tracks are NEXRAD storm attributes extrapolated in a straight line at constant speed.** Real cells turn, speed up, split and die; the arrival line is a rough heads-up, not a warning.
  - **Alerts are US-only.** `api.weather.gov` answers HTTP 400 outside its coverage, so the alerts request is skipped entirely unless the place is inside one of these boxes: lower 48 (lat 24-50, lon -125 to -66), Alaska (51-72, -180 to -129), Hawaii (18-23, -161 to -154), Puerto Rico / USVI (17-19, -68 to -64), Guam / CNMI (13-21, 144 to 146). They are boxes, not borders: a Canadian or Mexican place inside the lower-48 box still asks, and any HTTP error there reads as no alerts. Either way the alert strip just stays hidden.
  - **Radar refreshes only while the app is on screen.** Every 5 minutes, and at once on return if the last load is over 5 minutes old. A loop left in the background is not kept current until it is opened again.
  - **Radar timestamps are in the phone's timezone,** everything else is in the forecast location's local time. Only matters when looking at another time zone.
  - **Daily icon vs. daily precip %** can look inconsistent (an overcast icon next to 89%). Both come from Open-Meteo as-is: the icon is the day's dominant condition, the % is the single wettest hour. Not smoothed over.
  - **Days 8-10 are the loosest numbers on the screen.** Open-Meteo can return no precip % that far out, so a row can show a rain icon with no percentage.
  - Open-Meteo is a model blend, not a nowcast. Treat the hourly numbers as directionally right, not exact.
  - The Android app is **not on Google Play** and has no update mechanism — a new version means building a new APK and installing it over the old one. Since it is signed with the release key that install is now a plain in-place update; it was an uninstall-and-lose-your-data affair while the debug key was in use.
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## Web build files

|                                   |                                                            |
|-----------------------------------|------------------------------------------------------------|
| <code>index.html</code>           | markup                                                     |
| <code>app.css</code>              | all styling, dark theme, phone-first                       |
| <code>app.js</code>               | everything else — data, icons, render, radar, storm tracks |
| <code>sw.js</code>                | service worker, caches the shell only                      |
| <code>manifest.webmanifest</code> | PWA manifest                                               |
| <code>icons/</code>               | 192 / 512 / maskable-512 PNGs                              |
| <code>weather.apk</code>          | the published Android build                                |

Local preview:

```
python -m http.server 8765
```

Then open `http://localhost:8765`. A service worker is registered, so after editing `app.js` or `app.css`, hard-

reload (or bump `CACHE` in `sw.js`) to be sure you're seeing the new file.